

AI FAQ ASSISTANT API

Phase 4 : PROJECT PLANNING

Project Title: AI FAQ Assistant API

4. PROJECT PLANNING

Project planning organizes the work required to develop the AI FAQ Assistant API. It identifies objectives, scope, requirements, resources, tasks, schedule, milestones, team responsibilities, risks, testing, deployment, and maintenance. A structured plan helps the development team complete the project systematically and reduces the possibility of missing important activities.

The project is planned as a sequence of phases beginning with requirement analysis and continuing through design, data preparation, development, testing, deployment, and documentation. Each phase produces an output that becomes an input for the next phase.

4.1 Project Objectives

- Develop an API that accepts natural-language FAQ questions.
- Return relevant answers in a structured JSON format.
- Create a reusable service that can be integrated with different applications.
- Maintain FAQ information in a structured knowledge source.
- Apply input validation, security, logging, and error handling.
- Test the API with normal and abnormal inputs.
- Prepare complete technical and project documentation.

4.2 Project Scope

The scope includes requirement collection, system design, FAQ-data preparation, API development, question processing, retrieval or AI integration, testing, deployment preparation, and documentation. The initial system can focus on a selected FAQ domain so that the core functionality can be completed within the project schedule.

Features such as multilingual support, voice interaction, analytics, human escalation, continuous learning, and large-scale distributed deployment can be considered future enhancements. Clearly defining the scope helps the team concentrate on the required academic deliverables.

4.3 Functional Requirements

- The system shall accept questions through an API endpoint.
- The system shall validate incoming requests.
- The system shall search FAQ information or process questions through an AI component.
- The system shall return an answer in JSON format.
- The system shall return meaningful errors for invalid requests.
- Authorized users shall be able to manage FAQ records if administrative functions are implemented.

4.4 Non-Functional Requirements

- The API should provide consistent response formats.
- The application should be maintainable through modular code.
- The system should protect secret configuration values.
- The service should provide acceptable response time under normal workload.
- The system should be reliable and handle failures safely.
- The project should be scalable enough for its intended academic workload.

4.5 Technology Planning

Technology	Planned Use
Python	Main programming language.
FastAPI/Flask	API development framework.
JSON	Request and response format.
SQLite/PostgreSQL	FAQ data storage.
AI/NLP Service	Question understanding and answer

	generation.
Visual Studio Code	Development environment.
Git	Version control.
Postman/Swagger	API testing and documentation.

4.6 Resource Planning

Resource planning identifies the people, hardware, software, data, and external services required to complete the project. A normal computer or laptop with Python and internet access is sufficient for development. If an external AI service is used, an API key and suitable service access may be required.

- Human resources: analyst, developer, tester, documentation member, and coordinator.
- Hardware: computer/laptop and network connection.
- Software: Python, IDE, API framework, database, testing tools, and Git.
- Data: FAQ question-answer pairs, categories, keywords, and test cases.
- External services: optional AI API, cloud database, or hosting service.

4.7 Task Planning

Phase	Activities	Deliverable
1. Analysis	Collect requirements and define scope.	Requirement specification
2. Design	Prepare architecture, API, database, and data flow.	Design document
3. Data Preparation	Collect, clean, and categorize FAQ data.	FAQ dataset
4. Development	Build endpoints and	Working API

	processing modules.	
5. Testing	Execute test cases and correct defects.	Test report
6. Deployment	Configure and publish the application.	Deployment-ready service
7. Documentation	Prepare final technical documents.	Project documentation

4.8 Work Breakdown Structure

The Work Breakdown Structure divides the project into manageable work packages. The major packages are analysis, design, data preparation, development, testing, deployment, and documentation. Each package can be assigned to a team member and tracked using milestones.

- Analysis: identify users, FAQ domain, inputs, outputs, and constraints.
- Design: prepare system architecture, API endpoints, database and security design.
- Data: prepare FAQ records and test questions.
- Development: implement API, validation, retrieval, AI integration, and response handling.
- Testing: create test cases, execute them, record defects, and retest corrections.
- Deployment: configure environment, dependencies, secrets, and server.
- Documentation: prepare planning, design, usage, testing, and final reports.

4.9 Project Schedule

The following eight-week schedule is a sample academic project schedule. The actual dates may be adjusted according to the project submission deadline.

Week	Planned Activities
Week 1	Requirement analysis and project scope.

Week 2	System architecture and detailed design.
Week 3	FAQ dataset preparation and environment setup.
Week 4	API development and validation.
Week 5	FAQ retrieval and AI integration.
Week 6	Testing, debugging, and improvement.
Week 7	Deployment preparation and documentation.
Week 8	Final review, demonstration, and submission.

4.10 Milestones

- Requirements approved.
- System design completed.
- FAQ dataset prepared.
- First working API completed.
- AI or retrieval component integrated.
- Functional and integration testing completed.
- Deployment-ready version completed.
- Final documentation and demonstration completed.

4.11 Team and Role Planning

Role	Responsibilities
Project Coordinator	Coordinates tasks, meetings, schedule, and review.
System Analyst	Studies requirements and prepares

	specifications.
Backend Developer	Develops API endpoints and business logic.
AI/Data Developer	Prepares FAQ data and AI/NLP integration.
Tester	Creates test cases and reports defects.
Documentation Lead	Prepares project and technical documents.

In a small academic project, one person may perform more than one role. The role plan is therefore a responsibility guide rather than a requirement for a large team.

4.12 Risk Management

Risk	Impact	Mitigation
Poor FAQ data	Incorrect answers	Review and test the FAQ dataset.
AI service failure	Answer generation unavailable	Provide FAQ retrieval fallback.
API key exposure	Unauthorized use or cost	Store keys in environment variables.
Slow API response	Poor user experience	Optimize processing and limit input size.
Invalid input	Processing errors	Use validation and clear error responses.
Deployment failure	Service unavailable	Test deployment configuration early.
Schedule delay	Incomplete deliverables	Track milestones and prioritize core features.

4.13 Testing Plan

Testing is planned at multiple levels. Unit testing verifies individual functions. Integration testing checks communication between API, data storage, and AI components. System testing verifies the complete question-to-answer flow. Negative testing checks invalid input and failure conditions. The testing process should record test cases, expected results, actual results, and defects.

Test Area	Example
Valid Request	Submit a normal FAQ question and verify the response.
Empty Input	Submit an empty question and verify validation.
Invalid JSON	Send malformed JSON and verify controlled error.
Unknown Question	Verify fallback behavior for unsupported questions.
Authentication	Check restricted endpoints with invalid credentials.
Data Retrieval	Verify the relevant FAQ record is selected.
Health Check	Verify that the API reports service status.

4.14 Deployment Plan

Deployment begins by creating a clean environment and installing project dependencies. Configuration values and secret keys are provided through environment variables. The application is started using the selected framework's recommended server configuration. After startup, health and functional requests are sent to confirm that the API works correctly.

- Create or select the deployment environment.
- Install dependencies from the requirements file.

- Configure database and AI service settings.
- Set secret values securely.
- Start the API service.
- Run health and functional tests.
- Monitor logs and correct deployment issues.

4.15 Maintenance Plan

Maintenance includes correcting defects, updating FAQ information, monitoring performance, updating dependencies, and improving answer quality. FAQ content should be reviewed periodically so that obsolete information is removed. Software dependencies should also be updated carefully after testing.

The source code should be maintained in a version-control repository. Changes should be reviewed and tested before release. If an external AI service is used, changes to service limits, models, or configuration should be monitored and reflected in the project configuration.

4.16 Expected Outcome

The expected outcome is a working AI FAQ Assistant API that receives natural-language questions and returns useful answers through a documented interface. The completed system should demonstrate API development, AI-assisted question processing, data management, validation, testing, security awareness, and deployment practices.

4.17 Project Monitoring and Control

Project progress can be monitored by comparing completed tasks with the planned schedule. Each milestone should have a clear completion condition. Delays should be identified early and tasks should be rearranged when necessary. A simple weekly review can record completed work, pending work, problems, and the next week's priorities.

Review Item	Question
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Schedule	Are planned tasks completed on time?
Quality	Does the implementation meet requirements?
Testing	Are defects being identified and corrected?
Resources	Are required tools and data available?
Risk	Have new risks appeared?
Documentation	Are project documents updated?

4.18 Final Deliverables

- Requirement specification.
- Project design document.
- Working AI FAQ Assistant API source code.
- FAQ dataset or database.
- API testing results.
- Deployment configuration.
- User and technical documentation.
- Final project demonstration.

4.19 Summary

The project-planning document provides a structured roadmap for developing the AI FAQ Assistant API. It defines the objectives, scope, requirements, technologies, resources, work breakdown, schedule, milestones, roles, risks, testing, deployment, maintenance, monitoring, and final deliverables. Following this plan will help the team manage development in an organized manner and complete the project within the available academic schedule.